University of Galway Programming and Tools for Artificial Intelligence

(CT5132, CT5148, CT5229) 2026/2027 Assignments

James McDermott

This document describes continuous assessment (CA) for Programming and Tools for AI.

- **CT5132 and CT5148** have a written exam worth 50% of the module. R will not be examined in the written exam. It is "must-pass": a student who fails the exam, fails the module regardless of CA grade. CA is worth 50%. Of this Python is worth 45% and R 5%.
- **CT5229** has no written exam, so CA is worth 100%, all Python.

Deadlines

CT5132 and CT5148 have three submission deadlines, with exact times/dates announced in Canvas.

- End of Week 5: submit Python exercises of Weeks 0-4
- End of Week 10: submit Python exercises of Weeks 5-9
- End of Week 12: submit R assignment or R bootcamp certificate

CT5229 has two submission deadlines:

- End of Week 5: submit Python exercises of Weeks 0-4
- End of Week 12: submit Python exercises of Weeks 5-9

Python CA

We have 8 weeks of Python materials (Weeks 1-9, but Week 0 is just setup and Week 5 is a catch-up week). Each week contains a number of `.py` files containing exercises.

AI use

In these exercises I encourage students to use AI to **help you learn**, but not to use AI to **help you not learn** (Zvi Mowshowitz). The same goes for the solutions which are provided for most exercises. The same goes for looking up solutions on the internet. The same goes for copying solutions from your friends. Students should sometimes struggle with exercises before looking to these fallbacks, because often learning requires struggle. Students should submit their own solutions which they understand and can stand over.

Doctests, `selfcheck.py`, and submission format

Almost all exercises contain doctests, so students can check their own work.

There is a `selfcheck.py` script which can be run to confirm that the `.py` files are complete and pass the tests.

When ready to submit, ensure you have all your code in a directory, arranged like:

```
W01/E01_example.py
W01/E02_example.py
...
W09/E01_example.py
selfcheck.py
```

Run `python selfcheck.py --zip` to ensure your files are zipped correctly. The output is a zip for submission on Canvas. Don't create a zip by hand. If `python selfcheck.py --zip` doesn't create a zip, it's probably an older version of the script. Please use the version provided from Week 5+.

Interviews

We will hold an interview with every student. The purpose of the interview is to assess a realistic scenario in industry and in research: discussing code with colleagues.

Each interview will last approximately 10-20m. It will be conducted by the lecturer and/or the teaching assistant. For CT5132 it will be in-person by default; for CT5148 and CT5229 it will be online by default. We will make many interview time-slots available for booking by students. Please look for a separate announcement on this.

The interview will cover only the exercises where your submission includes an attempt at that exercise and it passes. So:

- Students are not incentivised to remove any incomplete or incorrect attempts.
- Students are incentivised to reflect on their own understanding before submission.
- Students are incentivised to remove solutions which they don't understand.

After interview, the interviewer will record brief notes and a number out of 10, where 0 indicates "did not show up for interview or was unable to answer any question", 5 indicates "some understanding, but substantial gaps" and 10 indicates "complete understanding".

The CA grade will be calculated as:

- The proportion of all exercises on which the submission passes the test, MULTIPLIED BY
- The proportion of understanding demonstrated at interview.

The interview aims to discuss code in a realistic way. We will ask questions like:

- What does this line do?
- Can you walk me through this algorithm?
- Why do we need this variable?
- What type is this value?
- What does this function return?
- What would happen if the caller passed-in such-and-such a value?
- Why did you give the variable that name?

- What is the time complexity of this function?

We will focus on exercises where the student's solution passed all doctests. We will try to ask some easy questions to start, and if the student answers easily, we will move to some harder ones. If a student struggles, we will allow time and try to rephrase questions to help them demonstrate some understanding.

We will not discuss grades during the interview. If we say "well done", "your answers are very good", or anything like that, it does not indicate the likely grade.

Optional earlier interview

We think some students will prefer to work ahead and get the interview finished early. If a student submits all their solutions early (before the CT5132/CT5148 Week 10 deadline, or before the CT5229 Week 12 deadline), they can do an early interview. We will make some interview slots available from approximately Week 7 onward to facilitate this.

R CA (CT5132, CT5148 only)

Most students in CT5148 will take an R bootcamp as part of another module, ST5001, and can deliver their R bootcamp certificate as the assessment for the R part of CT5148.

Any student not taking that bootcamp (likely including all CT5132 and possibly a few CT5148 students) should instead study the R materials provided as part of CT5132/CT5148 and submit a small R assignment. These materials and assignment will be released soon. There will be no interview in this.